

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/Static · 365 trials
Total training paths	1 · 365 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats → feature dim 1408
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=584 · test=73

Augmentation (training only)

Enabled	yes
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
SVM (RBF)	1.0000	0.0000	1.0000
SVM (Linear)	1.0000	0.0000	1.0000
Random Forest	1.0000	0.0000	1.0000
KNN	0.9931	0.0114	1.0000
Logistic Regression	1.0000	0.0000	1.0000
Voting Soft (SVM+RF+KNN+LogReg)	1.0000	0.0000	1.0000
Stacking (SVM+RF+KNN+LogReg)	1.0000	0.0000	1.0000
LDA	0.9470	0.0445	0.6027

Per-class hold-out F1

Class	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	Voting (soft)	Stacking	LDA	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.667	13
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.762	12
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.500	12
Double_Phone	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.381	12
Double_Pistol	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.571	12
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.692	12
macro avg	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.596	73

External test (NewTestData)

Algorithm	External test acc
Random Forest	1.0000
Logistic Regression	0.9355
Voting Soft (SVM+RF+KNN+LogReg)	0.9355
SVM (RBF)	0.9032
SVM (Linear)	0.9032
Stacking (SVM+RF+KNN+LogReg)	0.9032
KNN	0.8710
LDA	0.5484

Per-class external accuracy

[illegible]